

우주비행학

Home Work 4

Due date: May 24, 2018

- 1) Problem 2 in Chapter 3
- 2) Problem 4 in Chapter 3
- 3) Problem 1 in Chapter 7
- 4) Problem 4 in Chapter 7
- 5) Problem 5 in Chapter 7

3.10 PROBLEMS

1. A geosynchronous orbit has a period of $23^{\text{h}} 56^{\text{m}} 4.09^{\text{s}}$, or one sidereal day. Calculate the radius of a geosynchronous orbit. Starting from a parking orbit at $r_1 = 6578$ km and an inclination of 28° , calculate the two maneuvers required to reach geosynchronous, equatorial orbit. Do the required 28° inclination change at apogee as one burn by combining it with the second Hohmann transfer maneuver.
2. A lunar module (LM) lifts off from the lunar surface and flies a powered trajectory to its burnout point at 30 km altitude as shown in Figure 3.10. The velocity vector of the LM is parallel to the lunar surface at burnout. It then coasts halfway around the moon, where it must rendezvous with the Apollo command module (CM) in a 250-km circular orbit. The mass of the moon is 0.01213 times the mass of the earth, and the radius of the moon is 1740 km.
 - (a) What is the burnout speed of the LM? in kilometers per second?
 - (b) What is the magnitude of the maneuver required to rendezvous with the command module in kilometers per second?
 - (c) What is the coast time for the LM?
 - (d) In order to assure a quick rendezvous, it is desirable that the LM and CM arrive at the rendezvous point *together* (or at least close). Where must the CM be in relation to the LM at burnout? Cite your answer as either a time differential or angle differential of the CM ahead of or behind the LM burnout point. (This sets the "launch window" for the LM takeoff.)

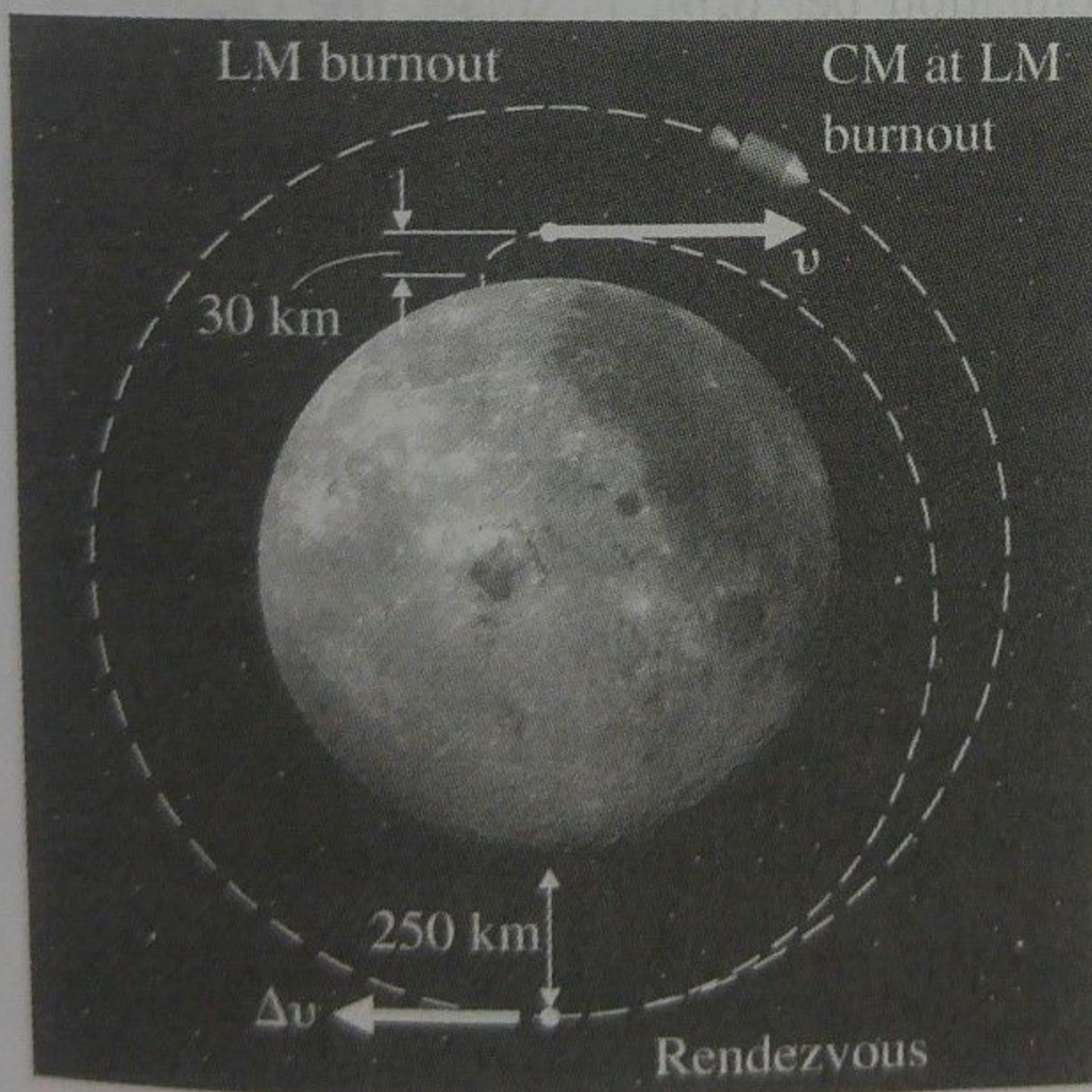


FIGURE 3.10

Lunar-module ascent trajectory.

3. On launch, the space shuttle does not directly inject into the final mission orbit. Rather, main-engine shutdown occurs at the apogee of an elliptical orbit at 100 km altitude. Perigee of this orbit is at only 30 km altitude, and the external tank is discarded to ensure that it will safely burn up over the open ocean. After the tank is dropped, the orbiter uses its orbital maneuvering system (OMS) to maneuver into another elliptical orbit with perigee at 100 km and apogee at 250 km altitude, the altitude

of the mission orbit. This is called the OMS 1 burn. At the new apogee, the shuttle performs the OMS 2 burn to circularize its orbit at the mission altitude. Calculate

- the burnout speed at apogee in the external tank disposal orbit,
 - the Δv required for the OMS 1 maneuver, and
 - the Δv required for the OMS 2 circularization maneuver.
- Do the last two maneuvers represent speed increases or speed decreases?

4. A low-thrust orbit-transfer vehicle is in a circular orbit and is attempting to change its orbital inclination i . The radius of the orbit is a , and the vehicle has a constant (low!) acceleration A . As shown in Figure 3.11, consider the thrust profile where

$$A = \begin{cases} -A\mathbf{w}, & -90^\circ < \nu < 90^\circ \\ +A\mathbf{w}, & 90^\circ < \nu < 270^\circ \end{cases} \quad (3.67)$$

where $\mathbf{w}$ is the orbit normal vector.

Write $\mathbf{r}$ in terms of the in-plane unit vectors $\mathbf{p}$, $\mathbf{q}$, and evaluate the specific torque on the orbit, $\mathbf{M} = \mathbf{r} \times \mathbf{A}$. Recall that $\mathbf{M} = d\mathbf{H}/dt$. As the thrust is low, integrate this over one orbit and show that the change in the angular-momentum vector is

$$\Delta\mathbf{H} = \int_0^T \mathbf{M} dt = 4Aa \sqrt{\frac{a^3}{\mu}} \mathbf{q} \quad (3.68)$$

(Hint: Show $dv/dt = (\mu/a^3)^{1/2}$.)

Argue that, since the change in inclination per orbit is small, the change in the inclination per orbit is

$$\Delta i \approx \frac{|\Delta\mathbf{H}|}{H} = \frac{4a^2 A}{\mu} \quad (3.69)$$

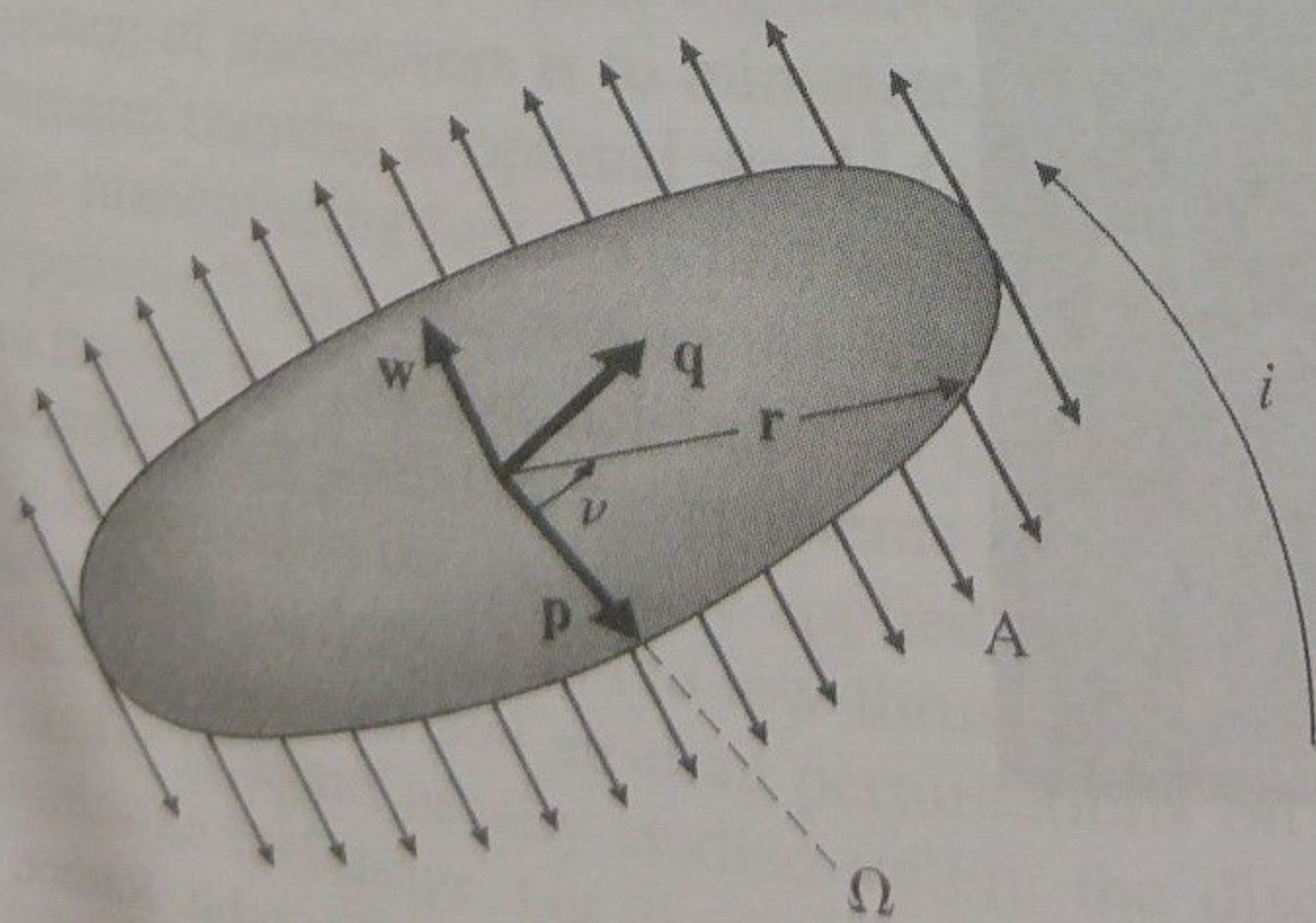


FIGURE 3.11
Low-thrust inclination change.

5. A satellite leaves a parking orbit at inclination i and executes a Hohmann transfer to geosynchronous equatorial orbit shown in Figure 3.12. Part of the required inclination change Δi_1 is performed during the first maneuver and the remainder $\Delta i_2 = i - \Delta i_1$

during the second maneuver. Show that the total inclination change is $\Delta i = \Delta i_1 + \Delta i_2 = i$.

Write the condition for the amount of inclination change (as in problem 1) is no



FIGURE 3.12
Hohmann transfer with

6. When the orbit of a satellite is called a ground track for a low-altitude circular



FIGURE 3.13
Ground-trace separation

7.14

PROBLEMS

1. The total Δv required for transfer from a low earth orbit at 28° inclination to geosynchronous equatorial orbit is $V_* = 4.29$ km/s. A space tug with a restartable hydrogen-oxygen engine has an $I_{sp} = 453$ s, $m_p = 16,000$ kg, and $m_s = 1300$ kg. How much payload can the tug deliver to geosynchronous orbit? Can the tug make a round trip without payload? If it can, how much payload could it carry to geosynchronous orbit and still return empty to low earth orbit?

2. Engine performance may sometimes be upgraded in a multistage rocket. Show that the trade-off ratio for engine improvement in stage k is given by

$$\left. \frac{\partial m_*}{\partial I_{sp,k}} \right|_{dV_*=0} = \frac{-g_c \ln(m_{0k}/m_{fk})}{\sum_{j=1}^N V_{ej} (1/m_{0j} - 1/m_{fj})} \quad (7.87)$$

Evaluate $\partial m_*/\partial I_{sp,0}$ and $\partial m_*/\partial I_{sp,1}$ for the space shuttle. Use the quantities pertaining to individual components (in parentheses) in Table 7.3. Should effort be placed into improving the solid-rocket boosters or the main engines first?

3. Calculate the trade-off ratios $\partial m_*/\partial m_{sk}$, $\partial m_*/\partial m_{pk}$, and $\partial m_*/\partial I_{sp,k}$ for the Von Braun shuttle. If the I_{sp} could be increased to 295 seconds in the upper two stages at the expense of a 7 percent increase in the structural mass of these stages, would the payload be increased or decreased? By how much?
4. For the sounding rocket in Table 7.5, calculate the approximate burnout speed (ignoring the effects of burn time) and maximum altitude achieved with a 250-kg payload. Suppose that instead of immediately igniting the second stage when the first burns out, we allow the second stage to coast to its maximum altitude and then ignite it. Calculate the maximum altitude achieved by the payload for this burn-coast-burn-coast strategy. Why is there such a large difference?

TABLE 7.5
Sounding rocket

k	I_{sp} , s	m_p , kg	m_s , kg
1	282	1167	113
2	282	415	41
*			250

5. The solid-rocket boosters of the space shuttle have an $I_{sp} = 290$ s, a dry mass of 81,900 kg each, and a full mass of 584,600 kg each. The external tank has a dry mass of 35,000 kg and contains 703,000 kg of propellant. Space-shuttle main engines have an $I_{sp} = 455$ s.
- (a) Consider a vehicle consisting of four solid-rocket boosters burning together as the first stage, and an external tank with two main engines as the second stage, as shown in Figure 7.13. Approximately how much payload can this vehicle deliver

- to low earth orbit ($V_* = 31,000$ ft/sec)? Set up the problem explicitly, and put in all numbers.
- (b) The mass of the payload can be ignored in some places. Where and why?
- (c) Calculate an approximate m_* to low earth orbit and compare to the shuttle payload (approximately 30,000 kg, or 65,000 lb).

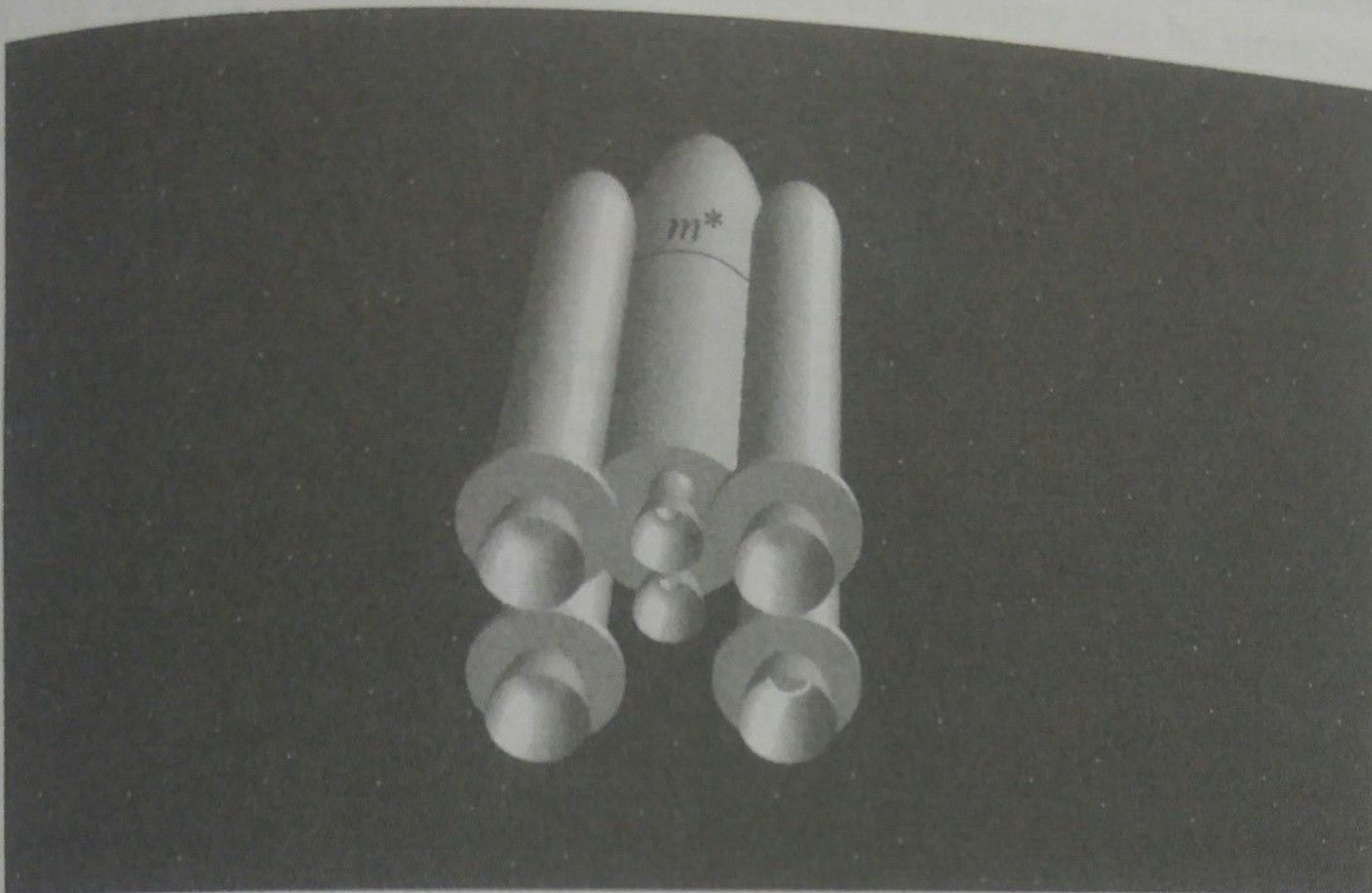


FIGURE 7.13

Shuttle-derived vehicle.

6. Using data on ϵ_k and π_k from the tables on the Von Braun three-stage rocket and the space shuttle, calculate values of λ for each stage. Is either vehicle close to an optimal design (that is, are the λ values nearly the same)?
7. Design a cargo third stage for the Von Braun three-stage rocket. Use the structural factor ϵ_2 from the second stage, since this stage does not have wings. What total payload could it deliver to orbit?